ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE D

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET second paper (practical, 80 marks): breadboard prototypes of a bridge-rectifier power supply, a two-door alarm with an OR gate IC, a temperature-controlled fan with a relay, and a relay latching circuit with its circuit diagram.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing for breadboard prototypes and draw.io (or similar) for circuit diagrams. If you do not have a computer, draw the breadboard layouts and circuit diagrams neatly by hand.
3. Put your name or examination number at the top of every printout or page.
4. Show the positive and negative supply rails clearly, place every component correctly and make all the connections.
5. Save your work regularly. In Fritzing, the 230 V side of the transformer may be shown as a plug or two labelled wires.

QUESTION 1

The circuit below is a full-wave bridge rectifier power supply with a smoothing capacitor and a power-on LED. Design and print its breadboard prototype.

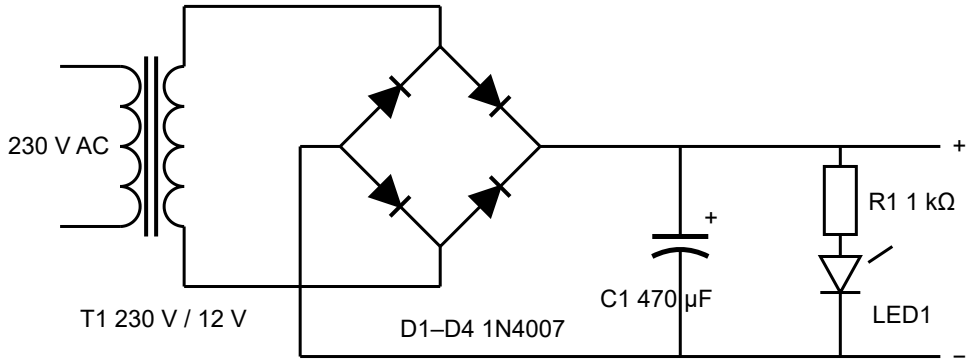


FIGURE 1: Bridge rectifier power supply

Components: 1 × 230 V / 12 V transformer (T1); 4 × 1N4007 diodes (D1–D4); 1 × 470 µF electrolytic capacitor (C1); 1 × 1 kΩ resistor (R1); 1 × green LED; breadboard and jumper wires.

Design and print (or draw) the breadboard prototype of the circuit in FIGURE 1. (20)

[20]

QUESTION 2

A robot workshop has two doors. A warning LED and a buzzer must come on when EITHER door is opened. Each door has a reed switch that closes when the door opens. The circuit uses one gate of a 74HC32 OR gate IC.

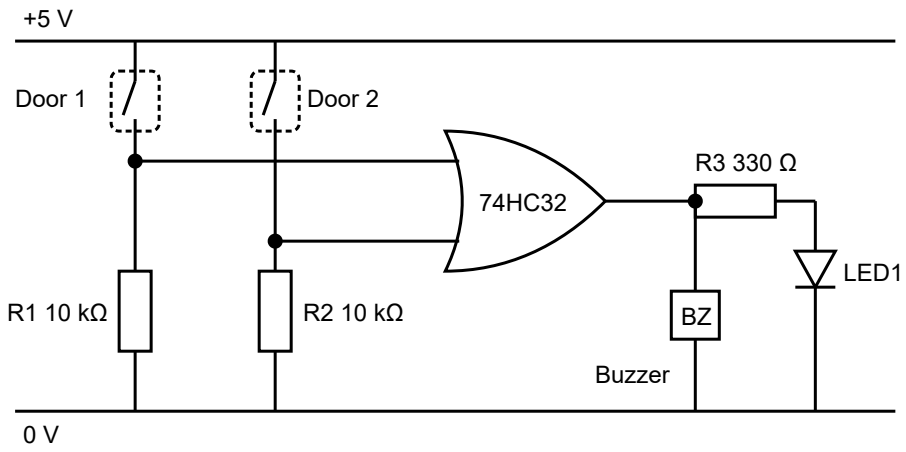


FIGURE 2: Two-door alarm (IC pin 14 to +5 V and pin 7 to 0 V are not shown)

Components: 1 × 5 V supply; 1 × 74HC32 quad OR gate IC; 2 × reed switches (or push buttons to represent them); 2 × 10 kΩ resistors (R1, R2); 1 × 330 Ω resistor (R3); 1 × red LED; 1 × 5 V active buzzer; breadboard and jumper wires.

Design and print (or draw) the breadboard prototype of the circuit in FIGURE 2. Use gate 1 of the IC (inputs on pins 1 and 2, output on pin 3) and remember the IC's supply pins. (20)

[20]

QUESTION 3

Thandi's robot controller box gets hot in summer. She wants a small 12 V fan to switch on automatically when the box gets warm. An NTC thermistor and a 10 kΩ potentiometer form a voltage divider at the base of a BC547 transistor. When the box heats up, the thermistor's resistance drops, the transistor switches on and energises a 12 V relay, whose contacts switch on the fan. A 1N4007 diode across the relay coil protects the transistor.

Components: 1 × 12 V DC supply; 1 × 10 kΩ NTC thermistor; 1 × 10 kΩ potentiometer; 1 × 1 kΩ resistor (base); 1 × BC547 NPN transistor; 1 × 12 V relay; 1 × 1N4007 diode; 1 × 12 V DC fan; breadboard and jumper wires.

Use Fritzing (or draw by hand) to design the breadboard prototype of the temperature-controlled fan. The thermistor must be at the top of the divider (to +12 V) and the potentiometer at the bottom (to 0 V). Add your name or examination number and print. (20)

[20]

QUESTION 4

Bongani needs a circuit that keeps a 12 V lamp on after a START button is pressed and released, until a STOP button is pressed. He uses a 12 V relay with two sets of normally open (NO) contacts: one set holds the relay on (latching) and the other switches the lamp. A diode across the relay coil absorbs the voltage spike when the relay switches off.

Components: 1 × 12 V DC supply; 1 × push button, normally open (START); 1 × push button, normally closed (STOP); 1 × 12 V relay with two NO contacts; 1 × 1N4007 diode; 1 × 12 V lamp; breadboard and jumper wires.

4.1 Draw the circuit diagram of the latching lamp circuit (in draw.io or neatly by hand), using the correct symbols. (5)

4.2 Design the breadboard prototype of the latching lamp circuit in Fritzing (or draw it by hand). Add your name or examination number and print. (15)

[20]

TOTAL: 80

END OF QUESTION PAPER